Principle Component Analysis

04 April 2025 20:45

(Principle Component Analysis)

* (9 – 11) ⟶ 15 mins (max)

* (Intro) ⟶ Rohit Jindal → Sr Data scientist (Retail Tech)

→ Non coding (DCE)/DTU (Civil Eng) ⟶ D.S

→ IRCON (2 years) [Level - 1 officer]

D.S → (Mastercard) → Paper Data Drift || SSGA (Financial M.L) || Target

→ IISC Bangalore * (Univ of Cambridge)

* (IP & Sustainability)

PCA → (simple)

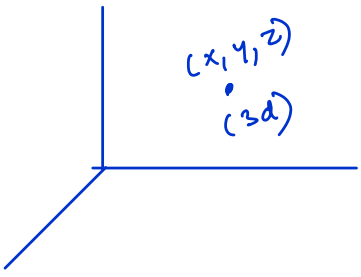
→ (Code)

→ optional* (Proof PCA*) → { quant / stats heavy }

What is PCA?

"d"

PCA is a technique to reduce the number of features (dimensions) in your data, while keeping as much information as possible.



⟹ * (Low dim space)

(some info will be lost)

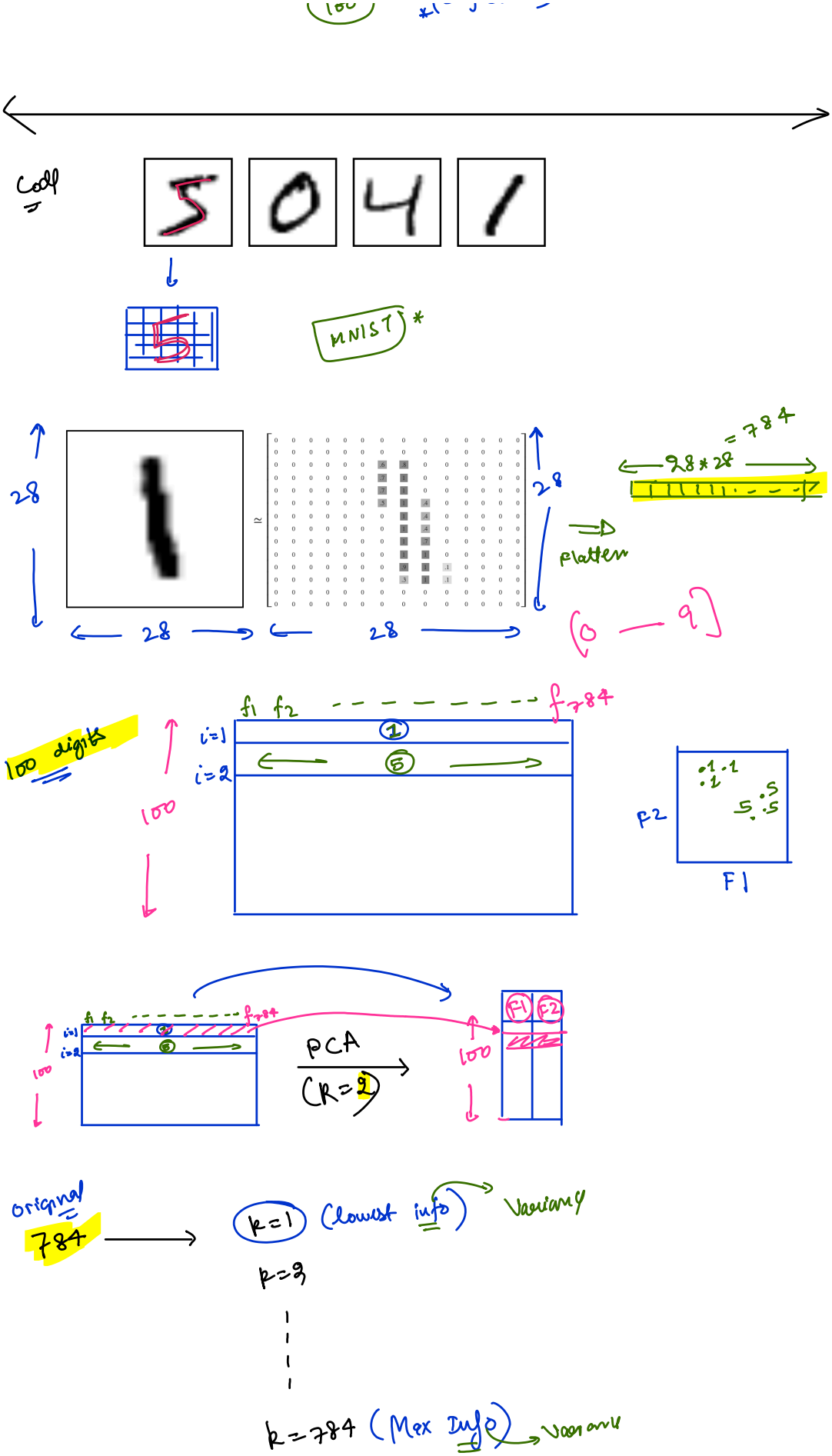
Why does PCA matter?

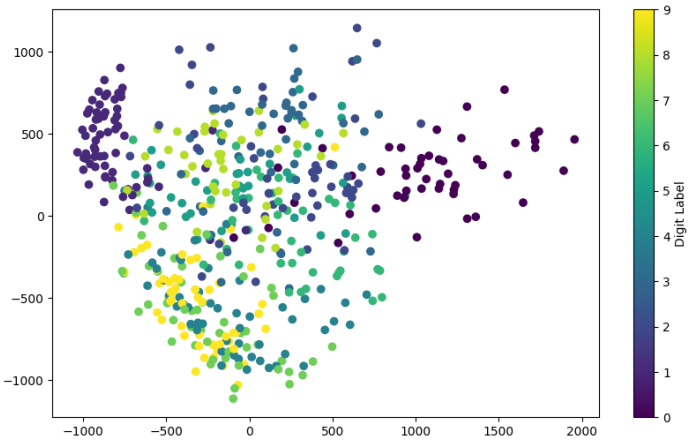
① Curse of Dim.

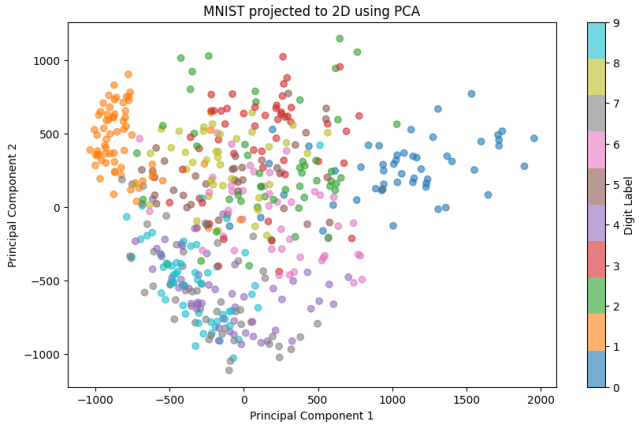
② Visualisation (challenge)

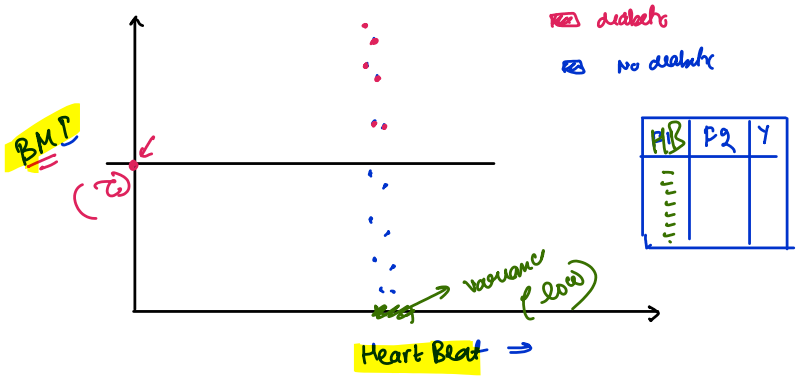
③ Speeding ML pipeline

→ (3 features)

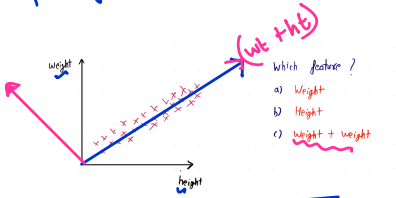

Coll
MNIST
28
28
28
28
= 784
28 * 28
flatten
(0 — 9)
100 digits
100
f1 f2
f784
i=1
i=2
F2
F1
PCA
(R=2)
original
784
k=1 (lowest info)
Variance
k=2
k=784 (Max Info)
Variance





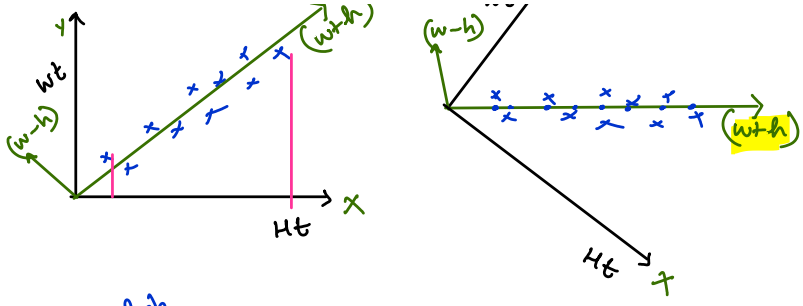


* feature where variance is high are more important



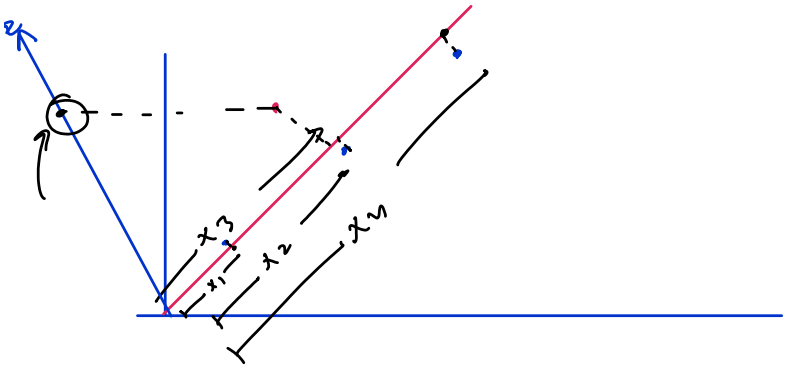
wt	Ht	Y





Plain English

Goal of PCA :→ we are trying to find such directions (in PCA) so that when we project our original data onto those directions, the variance of the projected data is maximised



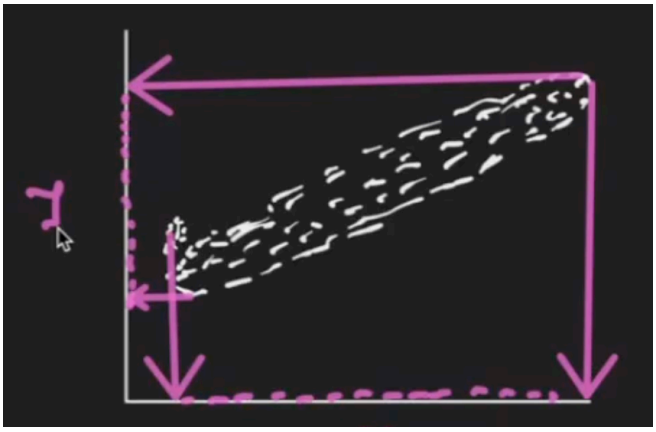


$$var = \frac{\sum (x_i - \bar{x})^2}{n}$$

X → Projection (scaler)

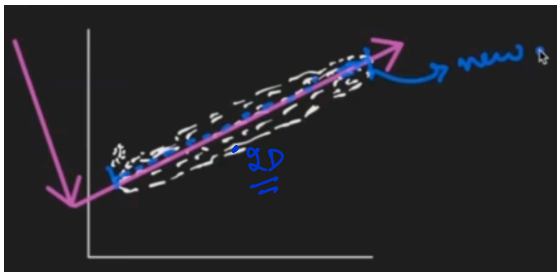
1 →	2
2 →	3
3	4
4	5
5	.
6	.
7	.
8	.
9	.
10	.

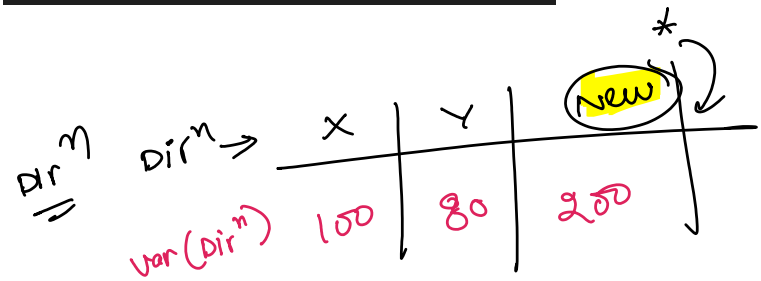
var(x) = 100

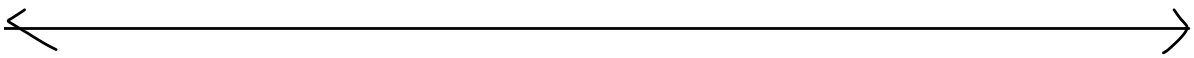


var(Y) = 8 units

var(New) = 200







Plain English

Goal of PCA :→ we are trying to find such w's directions (in PCA) so that when we project our original data onto those directions, the variance of the projected data is maximised

Setup $X \in R^{n \times d} \longrightarrow d' \ (d' < d)$

* $(d \longrightarrow 1)$

Lets find just one direction "w"

* (unknown dir^n)

Projection of X on w

$X_{(n \times d)} = \begin{bmatrix} & \end{bmatrix}$

* want? (centered) (var(Xw) is maximised) $w = \begin{bmatrix} \\ \end{bmatrix}$

want) Var(Xw) is maximised

$\frac{Xw - \bar{X}w}{N-1}$ → 0

$w(X - \bar{X})$

$\begin{bmatrix} \\ (-3.2) \end{bmatrix} =$

Variance =

X
2
3
4
5

$$\text{var} = \frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N-1}$$

$$\Rightarrow \text{var} = \frac{(2-3.2)^2 + (3-3.2)^2 + \cdots}{4-1}$$

$\Rightarrow Var(x^* w)$

x → already centered

$x - \bar{x} = 0$

(xw) → Projection

$\Rightarrow \max \left(Var(x^* w) \right)$ → variance of $\frac{(x_i - \bar{x})^2}{N}$

$= \max \left\{ \frac{\|xw\|^2}{N} \right\}$

$a^2 = a^T a$

$Var(xw) = \frac{(xw - \bar{x}w)^2}{N}$

$= \frac{(w(x - \bar{x}))^2}{N}$

$= (x^* w)^2$

$= \max \left\{ \frac{(xw)^T (xw)}{N} \right\}$

$\vec{a} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$(AB)^T = B^T A^T$

$a^2 = (2\ 3\ 4)\begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix}$

$= \max \left\{ \frac{w^T X^T X w}{N} \right\}$ → covariance matrix

$a^2 = a^T a = 2^2 + 3^2 + 4^2$

$= \max \left\{ w^T \Sigma w \right\}$

$\Sigma = \frac{X^T X}{N}$ *

$\Sigma = ?$

Height (cm)	Weight (kg)
160	55
170	65
180	75

$X =$

$Cov = \frac{X^T X}{N} = 2 \times 2$

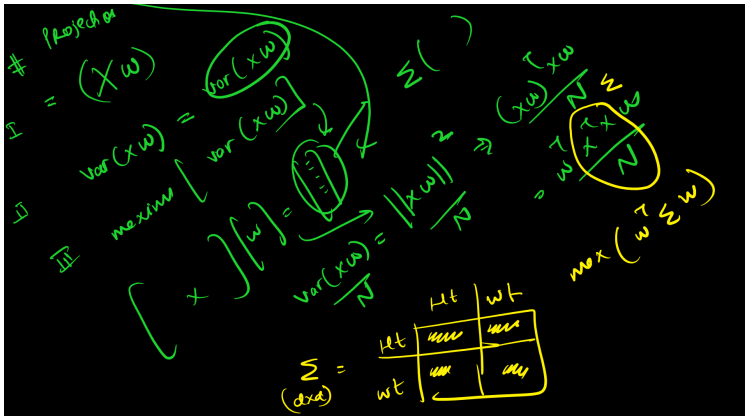
	Ht	wt
Ht	1	0.8
	0.8	1



Summary "w" → if I project my data, variance of projection is maximised (xw)

Maths $\max \{ w^T \Sigma w \}$ (Lagrangian Multiplier)

*(10:25 pm)



unsolvable

(#) $\max (w^T \Sigma w)$ { w → unknown (direction)

$\max (w^T \Sigma w)$

Constrained Problem

$\max (w^T \Sigma w)$ s.t $\|w\| = 1$

s.t $(\|w\| - 1 = 0)$

Let's **maximize:**

Lagrangian

$$f(x, y) = xy$$

Subject to:

$$g(x, y) = x + y - 10 = 0$$

Lagrangian

$$\mathcal{L}(x, y, \lambda) = f(x,y) - \lambda\, g(x,y) = 0$$

$$\mathcal{L} = xy - \lambda\,(x+y-10)$$

$$\frac{\partial \mathcal{L}}{\partial x} = 0 \quad \& \quad \frac{\partial \mathcal{L}}{\partial y} = 0 \qquad \Big| \; x = y = \lambda$$

$$\frac{\partial \mathcal{L}}{\partial x} \;\downarrow \quad y - \lambda(1) = 0$$

$$\frac{\partial \mathcal{L}}{\partial y} \;\downarrow \quad x - \lambda = 0$$

$$x = \lambda \quad -\text{I}$$

$$y = \lambda \quad -\text{II}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \qquad x + y - 10 = 0 \quad -\text{III}$$

Let's **maximize**:

$$f(x, y) = xy$$

Subject to:

$$g(x, y) = x + y - 10 = 0 \quad *$$

Step 1: Write the Lagrangian

$$\mathcal{L}(x, y, \lambda) = xy - \lambda(x + y - 10)$$

Step 2: Take partial derivatives

$$\frac{\partial \mathcal{L}}{\partial x} = y - \lambda = 0 \Rightarrow y = \lambda$$
$$\frac{\partial \mathcal{L}}{\partial y} = x - \lambda = 0 \Rightarrow x = \lambda$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x + y - 10) = 0 \Rightarrow x + y = 10$$

Step 3: Solve

From above:

- $x = y$
- $x + x = 10 \Rightarrow x = 5, y = 5$

original

$$\max\left(w^T \Sigma w\right) \quad \text{s.t} \quad \|w\| = 1$$

$$\max\left(w^T \Sigma w\right) \qquad \max\left(\lambda w^2\right) \qquad \max\left(\lambda\right)$$

$$\|w\| - 1 \geq 0$$

$$\|w\|^2 - 1 = 0 \Rightarrow \left(w^T w - 1 = 0\right)$$

$$w = \begin{bmatrix} \\ \\ \end{bmatrix} \qquad \Sigma = \begin{bmatrix} & \\ & \end{bmatrix}$$

$$\mathcal{L}_{(w,\lambda)} = w^T \Sigma w - \lambda\left(w^T w - 1\right)$$

$$w^T w = w^2 \;\Rightarrow\; 2w$$

(Matrix Derivatives)

$$\frac{\partial L}{\partial w} = 2\Sigma w - 2\lambda w = 0$$

$$\Rightarrow 2\Sigma w = 2\lambda w \Rightarrow \boxed{\Sigma w = \lambda w}$$

$w = [\ ]$ $\Sigma = [\ ]$ ($\lambda \rightarrow$ scalar)

condition

$\Sigma w = \lambda w$ → magical

$\max\left(w^T \Sigma w\right)$ s.t. $\|w\| = 1$

$w^T \lambda w = \lambda$ * (Pf)

$AX = \lambda X$

- Let their be a coloumn vector X and matrix A as given below :

$$X = \begin{bmatrix} -5 \\ -4 \\ 3 \end{bmatrix} \qquad A = \begin{bmatrix} 0 & 5 & -10 \\ 0 & 22 & 16 \\ 0 & -9 & -2 \end{bmatrix}$$

$$AX = \begin{bmatrix} 0 & 5 & -10 \\ 0 & 22 & 16 \\ 0 & -9 & -2 \end{bmatrix}_{3\times 3} \begin{bmatrix} -5 \\ -4 \\ 3 \end{bmatrix}_{3\times 1}$$

$$= \begin{bmatrix} 0 + (-20) + (-30) \\ 0 + (-88) + 48 \\ 0 + 36 + (-6) \end{bmatrix} = \begin{bmatrix} -50 \\ -40 \\ 30 \end{bmatrix} = AX$$

$\begin{bmatrix} -5 \end{bmatrix}$ *

$$AX = \begin{bmatrix} 10 & -4 \\ & 3 \end{bmatrix}$$

$AX = 10X$ *

$AX = \lambda X$ → (X → eigen vector of my matrix A)

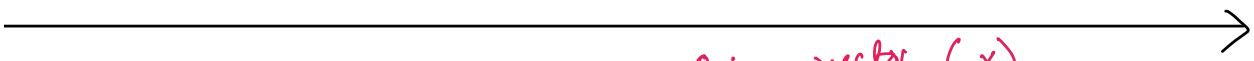
matrix $A_{3\times10}$ $X_{10\times1}$ = $(AX)_{3\times1}$

$$AX = \lambda X$$

"A" → (each matrix has some special eigen vectors)

(λ → corresponding eigen value)

covariance matrix → eigen vectors is that direction?



$$A = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$$

$AX = \lambda X$

$AX - \lambda X = 0$

$\Rightarrow (A - \lambda)X = 0$

Q. $A = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$; Find the Eigenvalues and Eigenvectors?

$\Rightarrow \det(A - \lambda I) = 0$

$$\det\left(\begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}\right) = 0$$

$$\det\begin{pmatrix} -5-\lambda & 2 \\ -7 & 4-\lambda \end{pmatrix} = 0$$

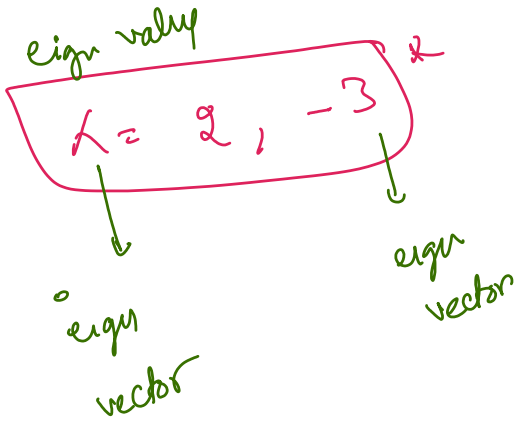
$$-(4-\lambda)(+5+\lambda) + 14 = 0$$

$\lambda = -3, +2 \rightarrow$ Eigenvalues

$\lambda_1 = +2, \lambda_2 = -3$

→ Taking det

$$|A - \lambda I| X = 0$$



$$AX = \lambda X$$

$$(\lambda - A) X = 0$$

$$X = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

---

eigen value $\lambda = 3, -11, 10$

vectors $\vec{X} = X_1 \quad X_2 \quad X_3$

$\left(\frac{10}{10+11+3} \times 100\%\right)$

(PCA) (3 features) → 1 (feature) ⇒ (X3)

→ 2 (features)

eigen value (λ) → amount of variance explained

Information loss

X — 5 dim (feature) —PCA→

5 eigen values

$\lambda_1 = 10 \rightarrow x_1$

$\lambda_2 = 7 \rightarrow x_2$

$\lambda_3 = 6 \rightarrow x_3$

$\lambda_4 = 3 \rightarrow x_4$

$\lambda_5 = 1 \rightarrow x_5$

explained variance Ratio → (5 dim → 2 dim)

(10, 7)

$$\text{Explained Variance} = \frac{10+7}{10+7+6+3+1} \times 100\%$$

$\Rightarrow$

$$\text{explained variance ratio} = \frac{10+7+6}{10+7+6+3+1} = \frac{23}{27}$$

$$= 85\%$$

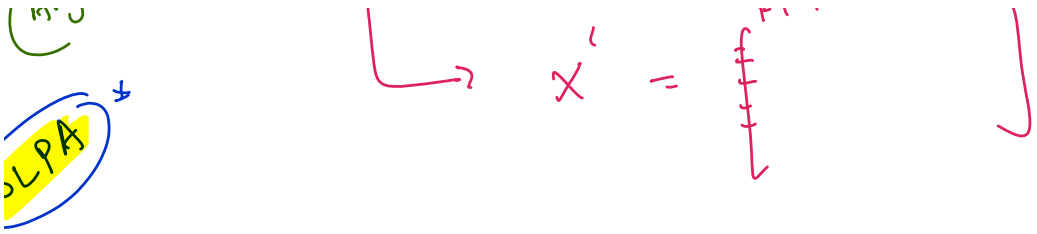
$\Rightarrow$ 85% of variance is preserved by top 3 PCs.

Summary

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_{100} \\ | & | & & | \end{bmatrix} \quad (k=5)$$

(Sum library)

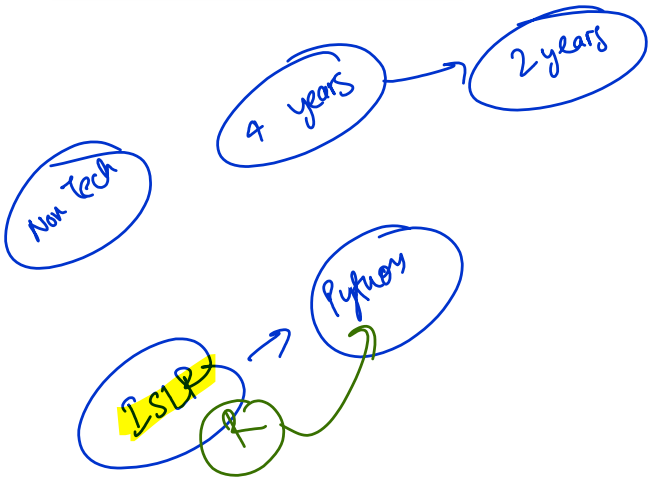
F1 F2 -- F5



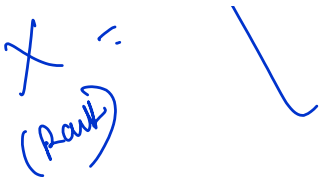
1. (Mean centers the data) = X centered
2. Find covariance matrix of X centered (Σ)
3. Find eigen value & e. vector of Σ
4. Take top k eigen values and top k eigen vectors
5. Project X onto those directions

0	0.46	0.16	0.18	-0.34
1	-1.54	-1.54	-0.72	0.96
2	0.16	0.66	0.08	-0.14
3	-0.14	-0.04	-0.12	-0.54
4	1.06	0.76	0.58	0.06

```
array([[ 0.55842085, -0.12653279],
       [-2.47546593,  0.23503028],
       [ 0.58090562, -0.06265197],
       [ 0.00869123, -0.56850538],
       [ 1.32744822,  0.52265986]])
```



(d) = no of eigen vectors





Agenda →

Robert Lindell
Sr D.S at Target

→ M.C / Steel Street

→ Civil engineer